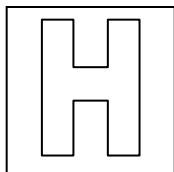


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PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2



CHEMISTRY

9647/01

Paper 1 Multiple Choice

23 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

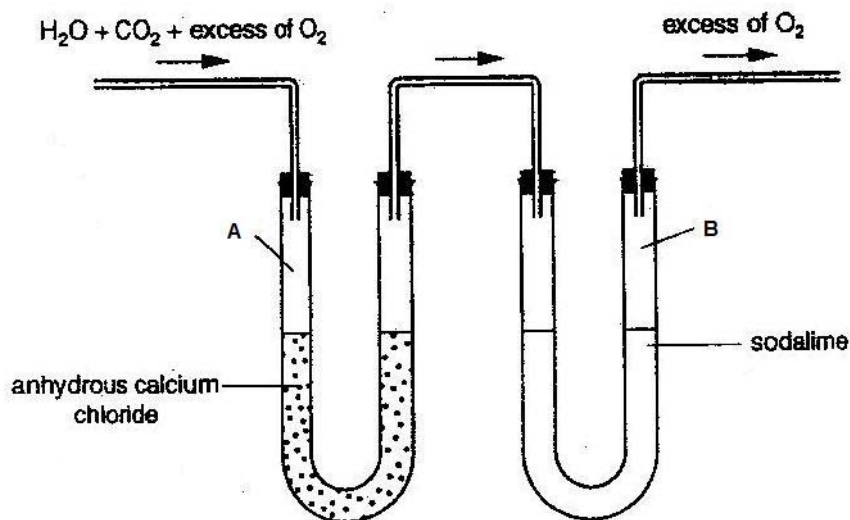
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 A sample of cyclohexene is completely burnt in dry oxygen and the gaseous products are collected as shown.



The increase in mass of the collecting vessels **A** and **B** of the apparatus are M_A and M_B respectively.

What is the ratio of M_A/M_B ?

- A 0.34
 - B 0.41
 - C 2.43
 - D 2.94
- 2 *The use of Data Booklet is relevant to this question.*

An ion E^{2+} contains 24 protons.

Which of the following statements about E^{2+} is **incorrect**?

- A The enthalpy change for the reaction $\text{E}(\text{g}) \rightarrow \text{E}^{2+}(\text{g}) + 2\text{e}$ is $+2243 \text{ kJ mol}^{-1}$.
- B The angle of deflection of E^{2+} in an electric field is smaller than that of E^{3+} .
- C The removal of the two electrons from **E** to form E^{2+} is from the 4s subshell.
- D E^{2+} is isoelectronic with Mn^{3+} .

3 Consider the following four compounds.

- 1 $(\text{CH}_3)_2\text{CHCO}_2\text{H}$
- 2 $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
- 4 $\text{CH}_3\text{CHFCH}_2\text{CH}_2\text{OH}$

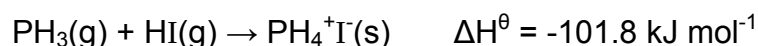
What is the order of decreasing volatility of the compounds?

- A $2 \rightarrow 1 \rightarrow 4 \rightarrow 3$
- B $2 \rightarrow 4 \rightarrow 1 \rightarrow 3$
- C $3 \rightarrow 1 \rightarrow 2 \rightarrow 4$
- D $4 \rightarrow 2 \rightarrow 3 \rightarrow 1$

4 Which of the following pairs of graphs share the same general shape according to ideal gas law for a fixed mass of gas?

- A V against T (at constant P) and PV against P (at constant T)
- B P against T (at constant V) and PV against P (at constant T)
- C P against V (at constant T) and V/T against T (at constant P)
- D PV against P (at constant T) and V/T against T (at constant P)

5 Phosphine reacts with hydrogen iodide to form phosphonium iodide in the reaction shown.



Given that $\Delta H_f^\ominus$ for $\text{PH}_3(\text{g}) = +5.4 \text{ kJ mol}^{-1}$ and $\Delta H_f^\ominus$ for $\text{HI}(\text{g}) = +26.5 \text{ kJ mol}^{-1}$, what is the standard enthalpy change of formation of phosphonium iodide?

- A $-133.7 \text{ kJ mol}^{-1}$
- B $-69.9 \text{ kJ mol}^{-1}$
- C $+69.9 \text{ kJ mol}^{-1}$
- D $+133.7 \text{ kJ mol}^{-1}$

6 In which of the following processes will ΔS be positive?

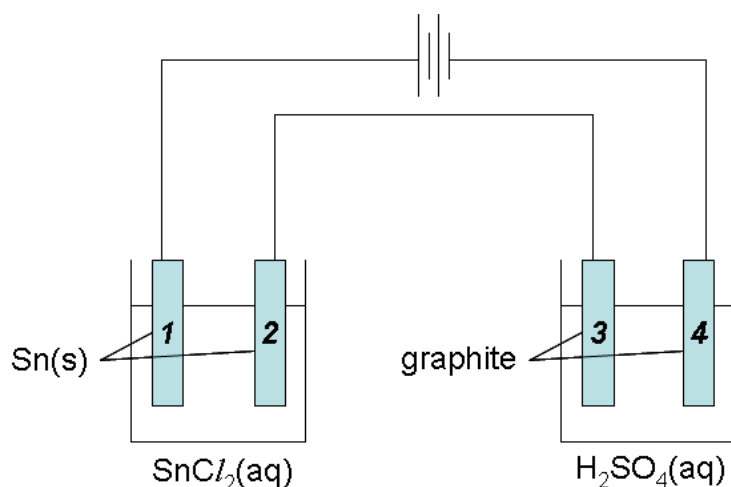
- A photosynthesis
- B electrolysis of water
- C condensation of steam
- D neutralisation of acid with alkali

- 7 A voltaic cell is made up of the Mg^{2+}/Mg half-cell and the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell.

Which one of the following statements is correct?

- A The Mg^{2+}/Mg half-cell is the positive electrode.
- B Increasing the temperature has no effect on the e.m.f. of the cell.
- C Addition of water to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ half-cell decreases the e.m.f. of the cell.
- D Addition of aqueous sodium hydroxide to the Mg^{2+}/Mg half-cell increases the e.m.f. of the cell.

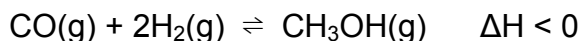
- 8 The following aqueous solutions were connected in series as shown below:



Which of the following correctly represents how the mass of each electrode changes as the reactions proceed?

	Electrode 1	Electrode 3	Electrode 4
A	no change	no change	decreases
B	decreases	no change	decreases
C	decreases	decreases	no change
D	no change	decreases	no change

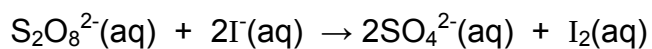
- 9 An equilibrium is represented by the following equation.



Which of the following changes would affect both the equilibrium constant K_p and the proportion of CH_3OH present at equilibrium?

- A adding a catalyst
 - B increasing the pressure
 - C reducing the temperature
 - D increasing the mass of CO
- 10 Which of the following statements about acids, bases and salts is correct?
- A 0.10 mol dm^{-3} of ethanoic acid has higher pH than 0.10 mol dm^{-3} of aqueous sodium ethanoate.
 - B 0.10 mol dm^{-3} of aqueous sodium ethanoate has higher pH than 0.10 mol dm^{-3} of aqueous ammonium chloride.
 - C 10 cm^3 of 0.10 mol dm^{-3} of hydrochloric acid reacts with 5 cm^3 of 0.10 mol dm^{-3} aqueous ammonia to produce an acidic buffer.
 - D 10 cm^3 of 0.10 mol dm^{-3} of ethanoic acid reacts with 5 cm^3 of 0.10 mol dm^{-3} sodium hydroxide to produce an alkaline buffer.

- 11** The kinetics of the reaction between iodide and peroxodisulfate can be investigated by varying the volume of the reactants used. The two reactants are mixed in the presence of a known amount of $\text{Na}_2\text{S}_2\text{O}_3$ and a little starch.



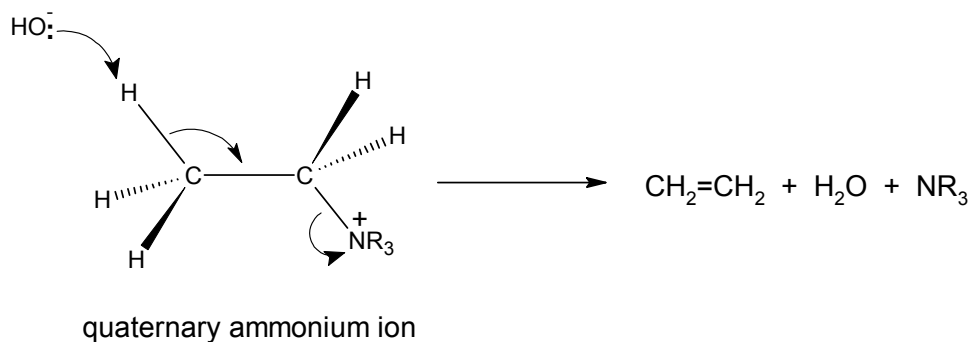
The time taken for an intense blue colour to be observed is then determined.

Experiment	Volume used / cm^3			Time / s
	1.0 mol dm^{-3} KI	0.040 mol dm^{-3} $\text{Na}_2\text{S}_2\text{O}_8$	H_2O	
1	10.0	5.0	25.0	170
2	15.0	5.0	20.0	113
3	15.0	10.0	15.0	56.5
4	20.0	x	y	21.3

What are the values of **x** and **y** in Experiment 4?

	x	y
A	5.0	15.0
B	10.0	10.0
C	15.0	5.0
D	20.0	0.0

- 12** Quaternary ammonium ions can undergo Hofmann elimination in a single step mechanism as shown.



The following concentrations of OH^- and $\text{CH}_3\text{CH}_2\text{NR}_3^+$ were used in *Experiment 1* and the half-life was found to be 10 s.

$$[\text{OH}^-] = 0.001 \text{ mol dm}^{-3}$$

$$[\text{CH}_3\text{CH}_2\text{NR}_3^+] = 0.100 \text{ mol dm}^{-3}$$

What will be the half-life when the experiment is repeated with the following changes in concentrations?

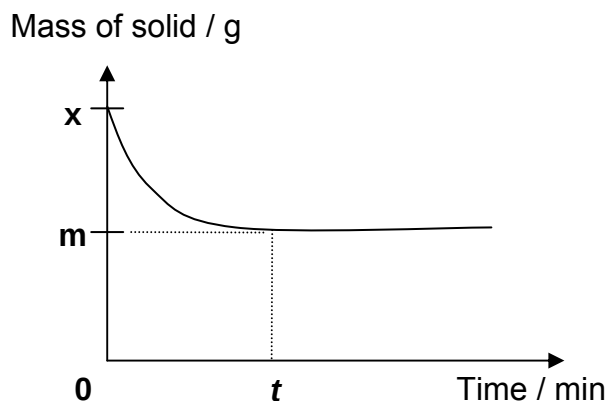
	Half-life for <i>Experiment 2</i> when $[\text{OH}^-]$ is doubled	Half-life for <i>Experiment 3</i> when $[\text{CH}_3\text{CH}_2\text{NR}_3^+]$ is doubled
A	10 s	10 s
B	10 s	5 s
C	5 s	10 s
D	5 s	5 s

- 13** *P*, *Q* and *R* are elements of the third period of the Periodic Table. The oxide of *P* is amphoteric, the oxide of *Q* is basic and oxide of *R* is acidic.

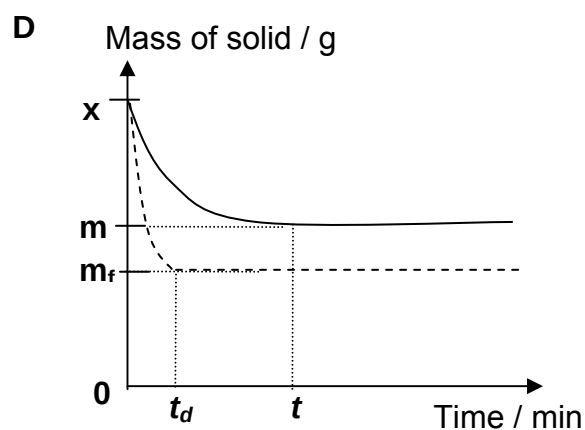
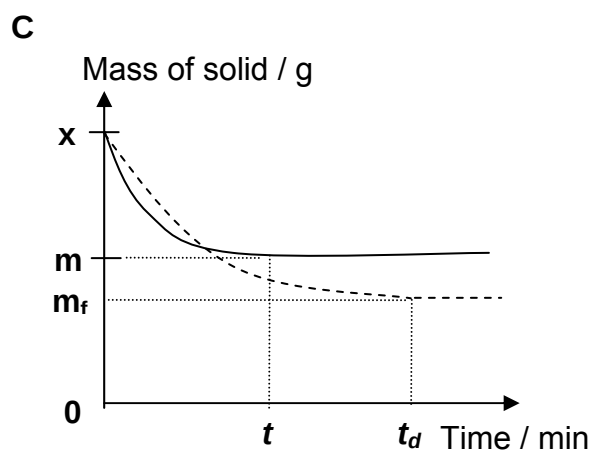
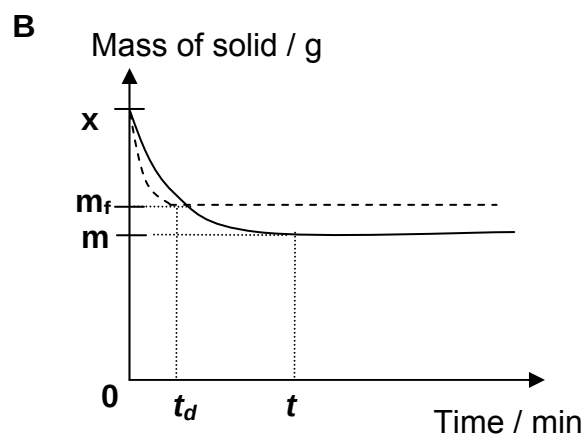
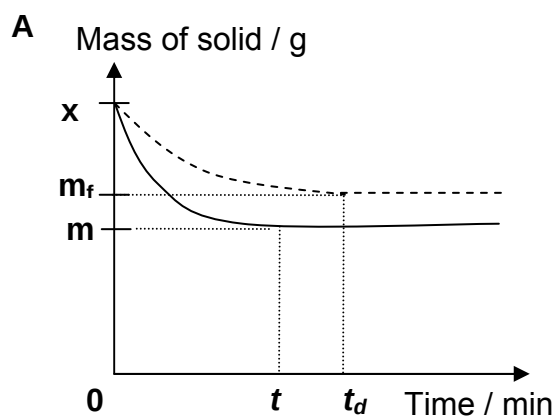
What is the order of increasing ionic radius?

- A** *RQP*
- B** *QPR*
- C** *PRQ*
- D** *PQR*

- 14 The graph below shows the variation in mass when x g of strontium nitrate is heated. At time t , decomposition is complete and the mass of the residue is m g.



Which of the following graphs below show the variation in mass when x g of magnesium nitrate is heated to a time t_d when the decomposition is complete with a final mass of m_f ?



- 15** When aqueous sodium chloride is added to aqueous silver nitrate, a white precipitate forms. The subsequent addition of aqueous sodium iodide to the mixture causes the white precipitate to dissolve and a yellow precipitate to form.

Which of the following helps to explain the observation?

- A** As a more powerful ligand, iodide ions can displace the chloride ions.
- B** The solubility product for silver iodide is lower than that for silver chloride.
- C** The magnitude of lattice energy of silver iodide is smaller than that of silver chloride.
- D** The covalent bonding between silver and iodine atom is stronger than that of silver and chlorine.

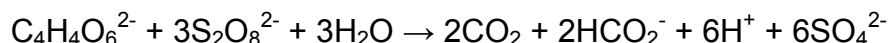
- 16** An aqueous solution containing a mixture of Al^{3+} , Fe^{2+} and Zn^{2+} ions is treated with an excess of aqueous ammonia.

What precipitate is left by this reaction?

- A** $\text{Al}(\text{OH})_3$ and $\text{Fe}(\text{OH})_2$
- B** $\text{Zn}(\text{OH})_2$ and $\text{Fe}(\text{OH})_2$
- C** $\text{Al}(\text{OH})_3$ and $\text{Zn}(\text{OH})_2$
- D** $\text{Fe}(\text{OH})_3$ only

- 17** *The use of Data Booklet is relevant to this question.*

Peroxodisulfate, $\text{S}_2\text{O}_8^{2-}$, is capable of oxidising the tartrate ion, $\text{C}_4\text{H}_4\text{O}_6^{2-}$, to carbon dioxide and methanoate as shown in the following reaction.



The reaction can be catalysed by a homogeneous catalyst. Given that the electrode potential for the tartrate ion is



Which metal ion is **not** a suitable catalyst for this reaction?

- A** V^{3+}
- B** Mn^{3+}
- C** Fe^{3+}
- D** Co^{3+}

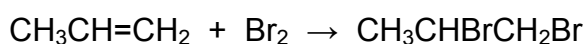
- 18** Methanoic acid, HCO_2H , is the first member of the homologous series of carboxylic acids, and occurs naturally in ants and in stinging nettles.

Unusually, in the presence of sulfuric acid and hydrofluoric acid, it adds to alkenes to form larger carboxylic acids.

If methanoic acid is added to but-1-ene, how many different isomeric acids (including structural and stereoisomers) are formed?

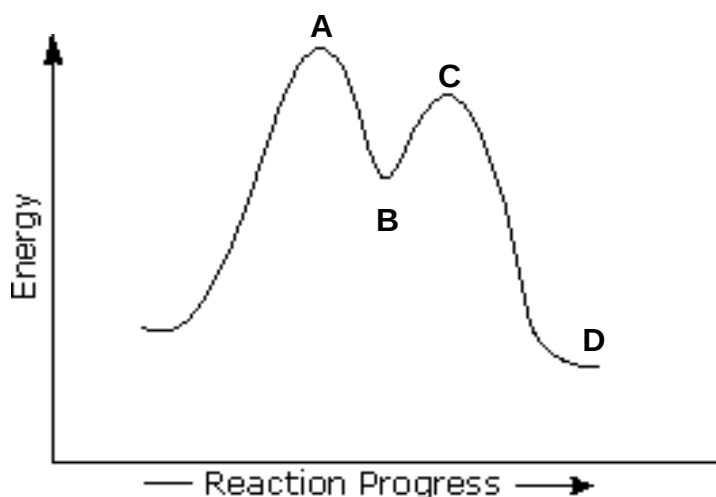
- A** 1 **B** 2 **C** 3 **D** 4

- 19** When Br_2 adds to propene



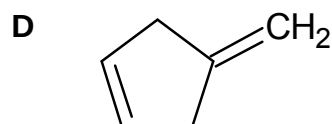
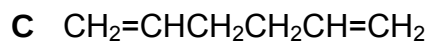
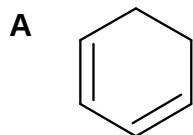
the reaction pathway diagram is as follows.

Which point in the diagram represents a positively charged intermediate?

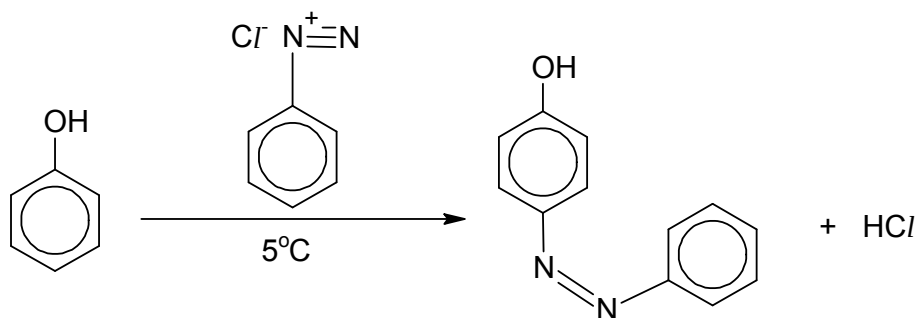


- 20** Compound **M** (C_6H_8) gives compound **N** ($\text{C}_4\text{H}_6\text{O}_4$) on oxidation. One mole of **M** reacts with two moles of hydrogen gas in the presence of platinum catalyst.

Which of the following shows the structure of **M**?



- 21 Phenol reacts with diazonium chloride as shown.



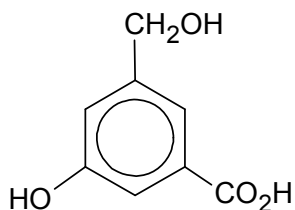
Which term describes this type of reaction?

- | | |
|--|--|
| <p>A nucleophilic substitution</p> <p>C electrophilic substitution</p> | <p>B nucleophilic addition</p> <p>D electrophilic addition</p> |
|--|--|
- 22 Chlorofluoroalkanes, commonly known as CFCs, undergo homolytic fission by ultraviolet irradiation in the stratosphere.

Which radical could result from this irradiation of $\text{CHFCl/CF}_2\text{Cl}$?

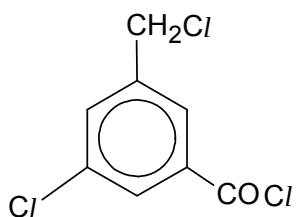
- A** $\text{CHFCl}\dot{\text{C}}\text{FCl}$
- B** $\dot{\text{C}}\text{HC/CF}_2\text{Cl}$
- C** $\dot{\text{C}}\text{HF CF}_2\text{Cl}$
- D** $\dot{\text{C}}\text{FC/CF}_2\text{Cl}$

- 23 An excess of phosphorus pentachloride is added to the compound below.

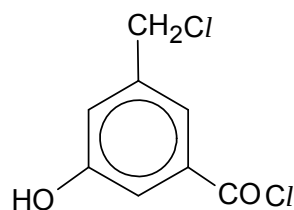


Which structure represents the organic compound formed?

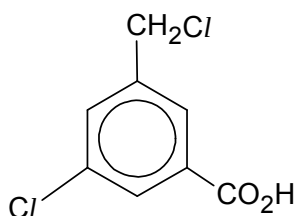
A



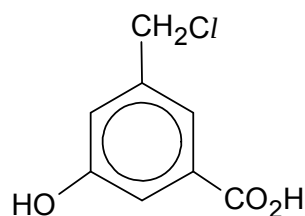
B



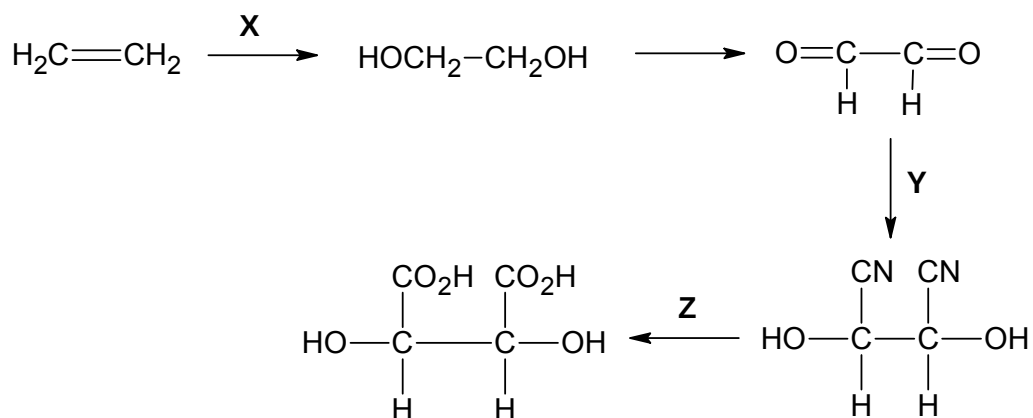
C



D



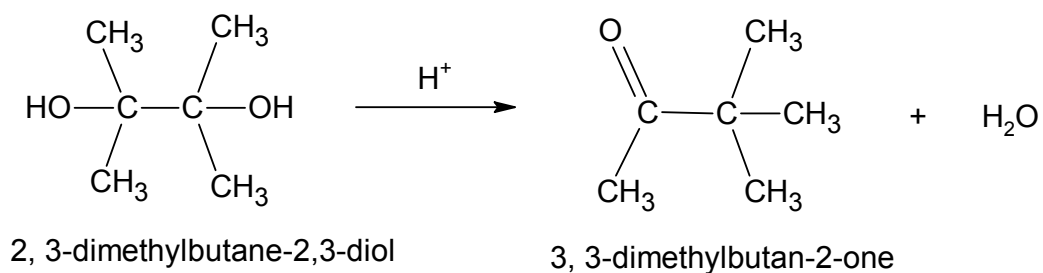
- 24 The following is a method of synthesising tartaric acid, a compound found in wine.



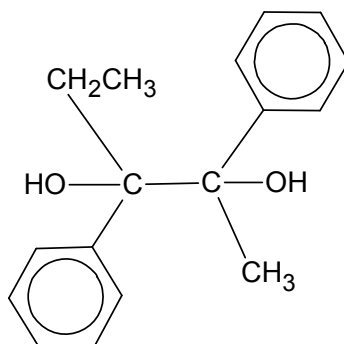
Which set of reagents and conditions can be used for the synthesis?

	stage X	stage Y	stage Z
A	cold KMnO_4 , NaOH(aq)	cold HCN , trace NaOH(aq)	hot $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4\text{(aq)}$
B	cold KMnO_4 , NaOH(aq)	ethanolic KCN , heat	HCl(aq) , heat
C	cold KMnO_4 , $\text{H}_2\text{SO}_4\text{(aq)}$	ethanolic KCN , heat	hot $\text{K}_2\text{Cr}_2\text{O}_7$, $\text{H}_2\text{SO}_4\text{(aq)}$
D	cold KMnO_4 , $\text{H}_2\text{SO}_4\text{(aq)}$	cold HCN , trace NaOH(aq)	HCl(aq) , heat

- 25 The pinacol rearrangement involves the reaction of a diol in acidic conditions to form a carbonyl compound as shown below.

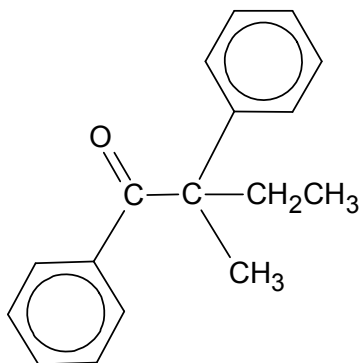


Which of the following structural formulae is **not** a product of the pinacol rearrangement of 2,3-diphenylpentane-2,3-diol with the structure as shown below?

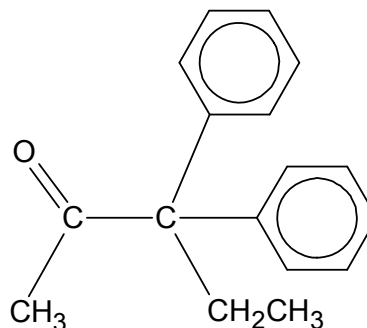


2,3-diphenylpentane-2,3-diol

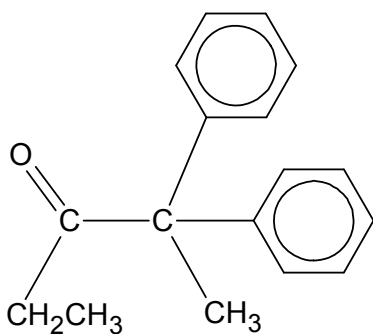
A



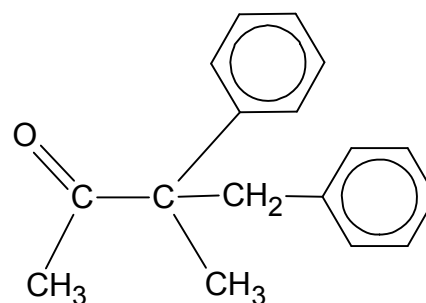
B



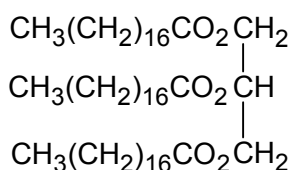
C



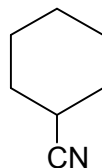
D



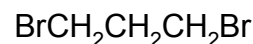
- 26 Which of the following does **not** yield ammonia and sodium propanoate when heated with aqueous sodium hydroxide?
- A $\text{CH}_3\text{CH}_2\text{CN}$
 B $\text{CH}_3\text{CH}_2\text{CONH}_2$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
 D $\text{CH}_3\text{CH}_2\text{CO}_2\text{NH}_4$
- 27 Which of the following reagents **cannot** be used to distinguish separate samples of methanoic acid and ethanal?
- A Tollens' reagent
 B alkaline aqueous iodine
 C 2,4-dinitrophenylhydrazine
 D aqueous sodium carbonate
- 28 Experiments are carried out on three compounds X, Y, and Z.



X



Y



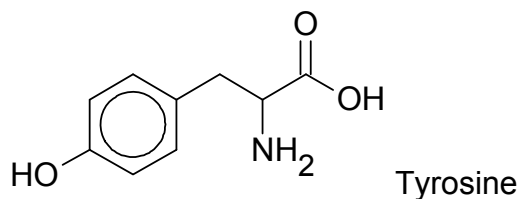
Z

A sample of 0.01 mol of each compound is heated under reflux with 50 cm³ of 1 mol dm⁻³ NaOH (in excess) until hydrolysis is complete and any ammonia produced is expelled from solution. The excess NaOH is then titrated in each case and is found to require 20 cm³, 30 cm³ and 40 cm³ of 1 mol dm⁻³ HCl for neutralisation.

Which sequence of compounds matches these results?

	20 cm ³	30 cm ³	40 cm ³
A	X	Y	Z
B	X	Z	Y
C	Y	Z	X
D	Z	Y	X

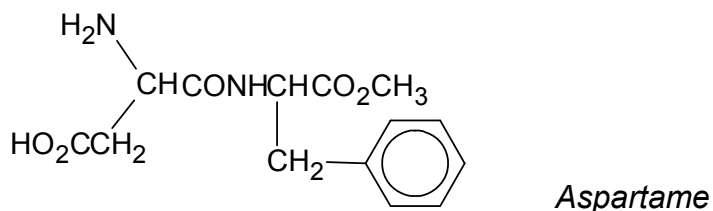
- 29 Tyrosine is one of the 20 amino acids that is used by cells to synthesise proteins, and was first discovered in cheese.



Which of the following types of reaction will Tyrosine **not** undergo?

- A reduction
- B nucleophilic addition
- C nucleophilic substitution
- D electrophilic substitution

- 30 *Aspartame* is widely used as a sweetener in canned drinks.



When *Aspartame* is treated with LiAlH_4 in dry ether, what is the final product obtained?

- A
- B
- C
- D

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

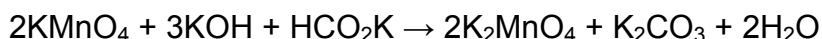
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31** A titration is carried out between 0.05 mol dm^{-3} potassium manganate(VII) in alkaline solution and 0.05 mol dm^{-3} potassium methanoate. The following reaction occurs.



Which of the following statements are correct about the reaction?

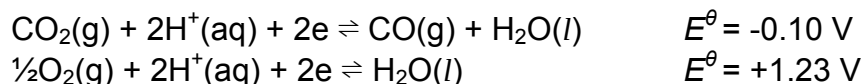
- 1** The potassium methanoate acts as a reducing agent.
 - 2** The volume of aqueous potassium methanoate solution required is half that of the aqueous potassium manganate(VII).
 - 3** The oxidation number of the carbon is increased by 2.
- 32** For which of the following pairs does the first ion have a smaller bond angle?
- 1** SO_4^{2-} , SO_3^{2-}
 - 2** NO_2^- , NO_3^-
 - 3** IF_4^- , BF_4^-

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 33** A domestic carbon monoxide detector is based on an electrochemical cell consisting of the electrodes:



Which statements about the cell are true?

- 1** In the cell, oxygen is reduced.
 - 2** Under standard conditions, the cell potential is +1.33 V.
 - 3** As the concentration of carbon monoxide in the detector increases, the cell potential becomes more positive.
- 34** The numerical values of the solubility product of calcium carbonate and calcium fluoride are 8.0×10^{-7} and 4.5×10^{-11} respectively at 30°C.

Which of the following statements are correct?

- 1** Calcium carbonate is more soluble than calcium fluoride.
 - 2** The solubility product of calcium carbonate decreases in a solution containing calcium nitrate.
 - 3** Addition of calcium nitrate solution to a solution containing equal concentration of carbonate and fluoride ions causes calcium carbonate to precipitate out first.
- 35** Tropeolin OOO indicator has a pH range of 7.6 to 9.0. Its colour is yellow at $\text{pH} < 7.6$ and rose-red at $\text{pH} > 9.0$.

Which of the following statements about Tropeolin OOO indicator are correct?

- 1** The indicator is yellow when 50 cm^3 of 0.20 mol dm^{-3} of aqueous ammonia is mixed with 50 cm^3 of 0.85 mol dm^{-3} aqueous ammonium chloride (pK_a of NH_4^+ is 9.25).
- 2** It is a suitable indicator for titration of ethanoic acid with sodium hydroxide.
- 3** The indicator is rose-red in a solution of 0.01 mol dm^{-3} of methylamine ($K_\text{b} = 4.37 \times 10^{-4} \text{ mol dm}^{-3}$).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 36** The cyanide ion CN^- closely resembles Cl^- in its chemical behaviour.

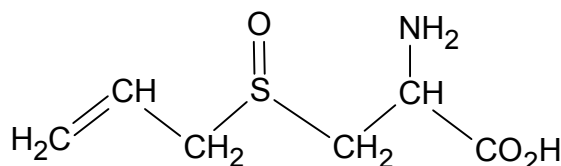
On this basis, which properties of compounds containing cyanide can be expected?

- 1 NaCN reacts with concentrated sulfuric acid to produce HCN gas.
- 2 HCN dissolves in water, ionises and forms an acidic solution.
- 3 AgCN is insoluble in water and aqueous ammonia.

- 37** Which of the following properties are identical for the two optical isomers of 2-hydroxypropanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$?

- 1 ΔH_f^θ
- 2 pK_a
- 3 melting point

- 38** Alliin is the compound responsible for the odour of garlic.



alliin

Which statements about this compound are correct?

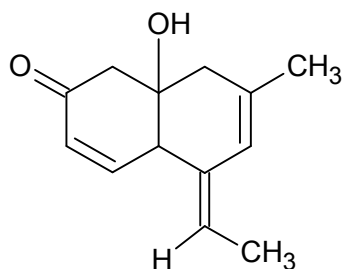
- 1 It can exist in optically active forms.
- 2 It can exist as zwitterions in aqueous solution.
- 3 It reacts with hot, acidified potassium dichromate(VI) to give carbon dioxide.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

39 Which of the following statements are correct about compound **H** shown below?



H

- 1** 1 mol of **H** reacts with 4 mol of hydrogen in presence of platinum catalyst.
- 2** 1 mol of **H** reacts with 1 mol of sodium to form 1 mol of hydrogen gas.
- 3** 1 mol of **H** reacts with 1 mol of sodium hydroxide.

40 The protein albumin (egg white) is precipitated from its aqueous solution when it is poured into **Z**.

What could **Z** be?

- 1** a saturated aqueous solution of $(\text{NH}_4)_2\text{SO}_4$
- 2** 0.1 mol dm^{-3} sulfuric acid
- 3** 0.1 mol dm^{-3} $\text{Pb}(\text{NO}_3)_2$

End of Paper

**Pioneer Junior College
2011 JC2 Preliminary Examination
H2 Chemistry Paper 1
Answers**

1	A	11	D	21	C	31	A
2	C	12	B	22	C	32	C
3	B	13	D	23	B	33	A
4	D	14	D	24	D	34	D
5	B	15	B	25	D	35	C
6	B	16	A	26	C	36	B
7	D	17	A	27	A	37	A
8	C	18	C	28	B	38	B
9	C	19	B	29	B	39	D
10	B	20	A	30	A	40	A